The Candy Factory Caper

Whodunnit? · Area, Surface Area & Volume Name: _____

At Sprinkle & Sons Candy Factory, the owner opened the safe to show visitors the **Golden Recipe**, the secret to the world's tastiest chocolate. The safe was open, and the recipe box was gone!

Six people were in the factory, each in a different place, and there are six ways to sneak a recipe out of a candy factory. You are the factory detective: find **who** took it, **where** they were, and **how** they got it out.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Mr. Sprinkle Factory Owner	☐ Chocolate River	☐ Packed in a candy tin
☐ Ms. Caramel Chocolatier	☐ Taffy Room	☐ Hidden in a gumball machine
☐ Gus Barton Forklift Driver	☐ Packing Floor	☐ Wrapped as a lollipop
☐ Dr. Fizz Flavor Scientist	☐ Gumdrop Garden	☐ Floated in chocolate
☐ Lolly Park Taste Tester	☐ Loading Dock	☐ Rolled in a candy barrel
☐ Inspector Mint Safety Inspector	☐ Test Kitchen	☐ Slipped in a sample bag

The culprit is _____ **in** _____ **using** _____

Show Your Work

The Candy Factory Caper · Show every step in the matching box.

Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Candy Factory Caper

Find the area of a triangle with base 10 cm and height 6 cm.

A. 15 cm²	The gumball machine was emptied onto a tray. Just red, blue, and green gumballs.
B. 16 cm²	Mr. Sprinkle was greeting guests in the Gumdrop Garden. The guest book shows the time.
C. 60 cm²	Dr. Fizz was taking a sample by the Chocolate River. A tour guide waved to her.
D. 30 cm²	Mr. Sprinkle was giving a tour by the Chocolate River. Twenty guests remember him.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Candy Factory Caper

Find the area of a trapezoid with bases 6 cm and 10 cm and height 4 cm.

A. 32 cm²	Every candy tin was opened at the dock. Only candy.
B. 24 cm²	Ms. Caramel was picking gumdrops in the Gumdrop Garden. Her basket was full of them.
C. 40 cm²	The coin slot on the gumball machine was jammed. Nothing could be dropped in.
D. 64 cm²	Dr. Fizz was testing taffy stretch in the Taffy Room. Ms. Caramel timed her.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Candy Factory Caper

Find the area of a circle with radius 3 cm. Leave your answer in terms of π.

A. 3π cm²	Gus Barton was parking his forklift near the Gumdrop Garden. The yard sensor logged it.
B. 9π cm²	Ms. Caramel was stretching taffy in the Taffy Room with her team.
C. 18π cm²	Every gumball machine was opened with the owner's key. They held only gumballs.
D. 6π cm²	Dr. Fizz was labeling boxes on the Packing Floor. The scanner log has her badge.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Candy Factory Caper

Find the volume of a rectangular prism that is 3 cm by 4 cm by 5 cm.

A. 12 cm³	Dr. Fizz was signing a delivery slip at the Loading Dock. The driver kept the copy.
B. 94 cm³	Inspector Mint was inspecting the hedges in the Gumdrop Garden. His clipboard has the time.
C. 60 cm³	The lollipops are all solid sugar, so nothing could be wrapped inside.
D. 20 cm³	The gumball machine in the lobby was weighed. It matched the gumball count exactly.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Candy Factory Caper

Find the surface area of a cube with edge length 5 cm.

A. 150 cm²	Gus Barton was loading trucks at the Loading Dock. The dock camera shows him.
B. 25 cm²	Lolly Park was sampling sour gumdrops in the Gumdrop Garden. The sample cup was empty.
C. 30 cm²	The gumball machine has a glass front and nothing was seen inside. No recipe in there.
D. 125 cm²	Dr. Fizz was mixing new flavors in the Test Kitchen. Lolly Park tasted each batch.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #6

The Candy Factory Caper

Find the volume of a cylinder with radius 2 cm and height 5 cm. Leave your answer in terms of π.

A. 50π cm³	Gus Barton was lifting pallets and passed by the Gumdrop Garden. A camera caught his forklift.
B. 80π cm³	The gumball machines were all wiped and checked. Only gumballs inside.
C. 10π cm³	Dr. Fizz was stirring a pot at the Chocolate River with a long paddle. Her sleeves are still sticky.
D. 20π cm³	The chocolate river filters showed nothing floating. No recipe came through.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Candy Factory Caper

Find the volume of a cone with radius 3 cm and height 4 cm. Leave your answer in terms of π.

A. 16π cm³	The gumball machine was shaken and tipped over. Not one thing but gumballs rolled out.
B. 12π cm³	Lolly Park was taste-testing mint chips in the Test Kitchen. The sign-in sheet shows it.
C. 4π cm³	Gus Barton was hauling a crate through the Gumdrop Garden when the bell rang. The sensor log shows it.
D. 36π cm³	Dr. Fizz was pulling sheets of taffy in the Taffy Room. The room's timer clip shows the minute.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

The Candy Factory Caper

Find the volume of a sphere with radius 3 cm. Leave your answer in terms of π.

A. 27π cm³	Mr. Sprinkle was watering the sugar-plants in the Gumdrop Garden. Visitors took pictures.
B. 108π cm³	Dr. Fizz was stacking cartons on the Packing Floor. A supervisor counted her boxes.
C. 36π cm³	The candy barrels were rolled and checked. They’re full of jelly beans.
D. 12π cm³	The gumball machine lock was sealed with tape. The tape was still intact.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

The Candy Factory Caper

Find the volume of a square pyramid with a 6 cm by 6 cm base and height 4 cm.

A. 144 cm³	Dr. Fizz was waving off the delivery truck at the Loading Dock. The gate log has the time.
B. 12 cm³	Ms. Caramel was dipping strawberries and then walked through the Gumdrop Garden. Her apron was spotless.
C. 48 cm³	Inspector Mint was checking labels on the Packing Floor, with a clipboard full of notes.
D. 72 cm³	A flashlight was shone into the gumball machine. It was all gumballs.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

The Candy Factory Caper

Find the surface area of a rectangular prism that is 3 cm by 4 cm by 5 cm.

A. 94 cm²	The sample bags were counted. All held only samples.
B. 60 cm²	Inspector Mint was measuring the paths of the Gumdrop Garden with a wheel. The wheel counter agrees.
C. 70 cm²	The gumball machine in the break room was opened. Only stale gumballs, nothing else.
D. 47 cm²	Dr. Fizz was timing a cookie bake in the Test Kitchen. The oven display shows the time.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.